

Distributional inverse drug retrieval for heterogeneous cellular populations

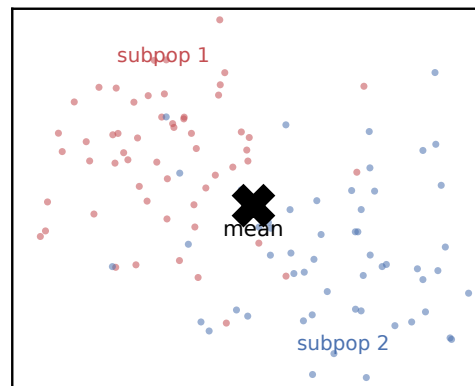
A Forward vs. inverse

Forward: drug + control → predicted states

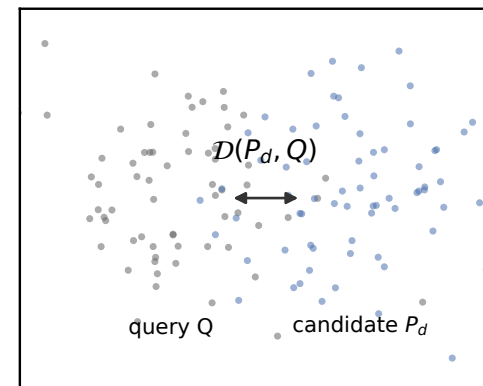
Inverse (this work): query population → ranked drugs

schematic

B Mean collapses subpopulations



C Distributional matching



D Ranking flip

Mean cosine picks

majority-biased drug ✗

Energy picks

covers-both ✓

→ a real drug decision changes

schematic (real flip in Fig. 2D)

E

Dataset ladder on the response-divergence axis

